

## **ASSIGNMENT-2**

### **Task – 1: System Utility Commands Practice**

#### **SOLUTION-**

##### **1.date-**

- The date command displays the current system date and time.

##### **2.uptime-**

- The uptime command shows how long the system has been running.

##### **3.hostname-**

- The hostname command displays the name of the computer on the network.

##### **4.which bash-**

- The which command locates the executable file of a command.

##### **5.uname -a**

- The uname -a command displays detailed system information.

##### **6.bc-**

- The bc is a command for line calculator used for mathematical calculations.

##### **7.cal-**

- The cal command displays a calendar in the terminal.

```
praveen@praveenasus: ~  
praveen@praveenasus:~$ date  
Sun Jun 7 09:55:00 UTC 2026  
praveen@praveenasus:~$ uptime  
09:55:04 up 7 min, 1 user, load average: 0.00, 0.02, 0.00  
praveen@praveenasus:~$ which bash  
/usr/bin/bash  
praveen@praveenasus:~$ hostname  
praveenasus  
praveen@praveenasus:~$ uname -a  
Linux praveenasus 6.6.114.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Mon Dec 1 20:46:23 UTC 2025 x86_64 GNU/Linux  
praveen@praveenasus:~$ bc  
bc 1.07.1  
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free  
Software Foundation, Inc.  
This is free software with ABSOLUTELY NO WARRANTY.  
For details type `warranty'.  
2+2  
4  
3*5  
15  
quit  
praveen@praveenasus:~$ cal  
Command 'cal' not found, did you mean:  
  command 'fcalf' from snap fcalf (4.0.6)  
  command 'col' from deb bsdxtrutils (2.41.3-3ubuntu2)  
  command 'cat' from deb coreutils-from-gnu (0.0.0~ubuntu25)  
  command 'cat' from deb coreutils-from-ubuntu (0.0.0~ubuntu25)  
  command 'cxl' from deb cxl (81-1ubuntu1)  
See 'snap info <snapname>' for additional versions.  
praveen@praveenasus:~$
```

## Task – 2: systemctl Deep Dive SOLUTION-

The systemctl command is used to manage services in Linux systems that use systemd. It allows administrators to start, stop, restart, enable, disable, and monitor system services efficiently.

Using **sudo apt install openssh-server & sudo apt install firewallld** install ssh and firewallld services for execute below commands.

### 1. Check the Status of the SSH Service Command:

systemctl status ssh

#### OUTPUT-

```
praveen@praveenasus:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; pres>
   Active: active (running) since Sun 2026-06-07 10:41:52 UTC; 12min >
  Invocation: 939b2817c04249099922a672326cbfda
 TriggeredBy: ● ssh.socket
    Docs: man:sshd(8)
          man:sshd_config(5)
   Main PID: 2208 (sshd)
     Tasks: 1 (limit: 4333)
    Memory: 1.6M (peak: 3.8M)
       CPU: 58ms
    CGroup: /system.slice/ssh.service
            └─2208 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 sta>

Jun 07 10:41:52 praveenasus systemd[1]: Starting ssh.service - OpenBSD >
Jun 07 10:41:52 praveenasus sshd[2208]: Server listening on 0.0.0.0 por>
Jun 07 10:41:52 praveenasus sshd[2208]: Server listening on :: port 22.
Jun 07 10:41:52 praveenasus systemd[1]: Started ssh.service - OpenBSD S>
```

The systemctl status ssh command displays the current status of the SSH service. **2. Start**

### the Firewall Service

#### Command:

sudo systemctl start firewallld-

This command starts the firewall service immediately.

### 3. Stop the Firewall Service

#### Command:

sudo systemctl stop firewallld-

This command stops the firewall service.

```
praveen@praveenasus: ~  
praveen@praveenasus:~$ sudo systemctl start firewalld  
praveen@praveenasus:~$ sudo systemctl status firewalld  
● firewalld.service - firewalld - dynamic firewall daemon  
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; vendor preset: enabled)  
   Active: active (running) since Sun 2026-06-07 10:54:18 UTC; 12s ago  
     Invocation: ff7eadc5bda440b88efe00de74dac4f3  
       Docs: man:firewalld(1)  
    Process: 3237 ExecStartPost=/usr/bin/firewall-cmd --state (code=exited, status=0/SUCCESS)  
   Main PID: 3236 (firewalld)  
      Tasks: 2 (limit: 4333)  
     Memory: 33.2M (peak: 59.5M)  
        CPU: 875ms  
    CGroup: /system.slice/firewalld.service  
            └─3236 /usr/bin/python3 /usr/sbin/firewalld --nofork --nopoll  
  
Jun 07 10:54:17 praveenasus systemd[1]: Starting firewalld.service - firewalld  
Jun 07 10:54:18 praveenasus systemd[1]: Started firewalld.service - firewalld  
praveen@praveenasus:~$ sudo systemctl stop firewalld  
praveen@praveenasus:~$ sudo systemctl status firewalld  
● firewalld.service - firewalld - dynamic firewall daemon  
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; vendor preset: enabled)  
   Active: inactive (dead)  
     Docs: man:firewalld(1)  
  
Jun 07 10:50:43 praveenasus systemd[1]: Starting firewalld.service - firewalld  
Jun 07 10:50:43 praveenasus systemd[1]: Started firewalld.service - firewalld  
Jun 07 10:52:38 praveenasus systemd[1]: Stopping firewalld.service - firewalld  
Jun 07 10:52:38 praveenasus systemd[1]: firewalld.service: Deactivated successfully  
Jun 07 10:52:38 praveenasus systemd[1]: Stopped firewalld.service - firewalld  
Jun 07 10:54:17 praveenasus systemd[1]: Starting firewalld.service - firewalld  
Jun 07 10:54:18 praveenasus systemd[1]: Started firewalld.service - firewalld  
Jun 07 10:56:28 praveenasus systemd[1]: Stopping firewalld.service - firewalld  
Jun 07 10:56:28 praveenasus systemd[1]: firewalld.service: Deactivated successfully  
Jun 07 10:56:28 praveenasus systemd[1]: Stopped firewalld.service - firewalld  
lines 1-15/15 (END)
```

#### 4. Enable a Service at Boot

##### Command:

```
sudo systemctl enable sshd
```

The enable option configures the SSH service to start automatically whenever the system boots.



```
praveen@praveenasus:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/syst
emd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
praveen@praveenasus:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: en>
   Active: active (running) since Sun 2026-06-07 10:41:52 UTC; 16min ago
   Invocation: 939b2817c04249099922a672326cbfda
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 2208 (sshd)
     Tasks: 1 (limit: 4333)
    Memory: 1.6M (peak: 3.8M)
       CPU: 58ms
    CGroup: /system.slice/ssh.service
            └─2208 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jun 07 10:41:52 praveenasus systemd[1]: Starting ssh.service - OpenBSD Secure>
Jun 07 10:41:52 praveenasus sshd[2208]: Server listening on 0.0.0.0 port 22.
Jun 07 10:41:52 praveenasus sshd[2208]: Server listening on :: port 22.
Jun 07 10:41:52 praveenasus systemd[1]: Started ssh.service - OpenBSD Secure >
lines 1-18/18 (END)
```

## 5. List All Running Services

### Command:

`systemctl list-units --type=service --state=running-`

This command lists all services that are currently running on the system.

```
praveen@praveenasus:~$ sudo systemctl list-units --type=service --state=running
```

| UNIT                        | LOAD   | ACTIVE | SUB     | DESCRIPTION                  |
|-----------------------------|--------|--------|---------|------------------------------|
| chrony.service              | loaded | active | running | chrony, an NTP client/server |
| cron.service                | loaded | active | running | Regular background progra    |
| dbus.service                | loaded | active | running | D-Bus System Message Bus     |
| getty@tty1.service          | loaded | active | running | Getty on tty1                |
| networkd-dispatcher.service | loaded | active | running | Dispatcher daemon for sys    |
| polkit.service              | loaded | active | running | Authorization Manager        |
| rsyslog.service             | loaded | active | running | System Logging Service       |
| ssh.service                 | loaded | active | running | OpenBSD Secure Shell serv    |
| systemd-journald.service    | loaded | active | running | Journal Service              |
| systemd-logind.service      | loaded | active | running | User Login Management        |
| systemd-resolved.service    | loaded | active | running | Network Name Resolution      |
| systemd-udevd.service       | loaded | active | running | Rule-based Manager for De    |
| unattended-upgrades.service | loaded | active | running | Unattended Upgrades Shutd    |
| user@1000.service           | loaded | active | running | User Manager for UID 1000    |
| wsl-pro.service             | loaded | active | running | Bridge to Ubuntu Pro agen    |

Legend: LOAD → Reflects whether the unit definition was properly loaded.  
ACTIVE → The high-level unit activation state, i.e. generalization of  
SUB → The low-level unit activation state, values depend on unit t

15 loaded units listed.  
lines 1-22/22 (END)

### Task – 3: ps Command Exploration

#### SOLUTION-

##### 1. ps

The ps command displays the processes running in the current terminal session only

##### 2. ps -e

The -e option displays all processes currently running on the system, including both system and user processes.

##### 3. ps aux

The ps aux command displays detailed information about all processes.

##### 4. ps -ef

The ps -ef command displays all processes in full-format listing.

## 5. ps -u root

The `ps -u root` command displays only the processes owned by the root user.

```
praveen@praveenasus:~$ ps
  PID TTY          TIME CMD
  336 pts/0      00:00:00 bash
  483 pts/0      00:00:00 ps
praveen@praveenasus:~$ ps -e
  PID TTY          TIME CMD
    1 ?           00:00:01 systemd
    2 hvcd0      00:00:00 init-systemd(Ub
    6 hvcd0      00:00:00 init
   50 ?           00:00:00 systemd-journal
   85 ?           00:00:00 systemd-resolve
   92 ?           00:00:00 systemd-udevd
  150 ?           00:00:00 chronyd-starter
  151 ?           00:00:00 cron
  152 ?           00:00:00 dbus-daemon
  156 ?           00:00:00 networkd-dispat
  161 ?           00:00:00 systemd-logind
  165 ?           00:00:00 wsl-pro-service
  181 ?           00:00:00 rsyslogd
  183 ?           00:00:00 sshd
  213 ?           00:00:00 chronyd
  227 ?           00:00:00 chronyd
  231 ?           00:00:00 unattended-upgr
  240 tty1       00:00:00 agetty
  334 ?           00:00:00 SessionLeader
  335 ?           00:00:00 Relay(336)
  336 pts/0      00:00:00 bash
  338 ?           00:00:00 login
  383 ?           00:00:00 systemd
  385 ?           00:00:00 (sd-pam)
  411 pts/1      00:00:00 bash
  492 ?           00:00:00 (udev-worker)
  493 ?           00:00:00 (udev-worker)
  499 pts/0      00:00:00 ps
praveen@praveenasus:~$
```

praveen@praveenasus: ~

praveen@praveenasus:~\$ ps aux

| USER     | PID | %CPU | %MEM | VSZ     | RSS   | TTY   | STAT | START | TIME | COMMAND          |
|----------|-----|------|------|---------|-------|-------|------|-------|------|------------------|
| root     | 1   | 1.0  | 0.4  | 24560   | 15064 | ?     | Ss   | 16:09 | 0:01 | /sbin/init       |
| root     | 2   | 0.0  | 0.0  | 3172    | 2212  | hvc0  | Sl+  | 16:09 | 0:00 | /init            |
| root     | 6   | 0.0  | 0.0  | 3172    | 2048  | hvc0  | Sl+  | 16:09 | 0:00 | plan9 --contr    |
| root     | 50  | 0.2  | 0.4  | 42184   | 16076 | ?     | S<s  | 16:09 | 0:00 | /usr/lib/syst    |
| systemd+ | 85  | 0.1  | 0.3  | 22524   | 14260 | ?     | Ss   | 16:09 | 0:00 | /usr/lib/syst    |
| root     | 92  | 0.4  | 0.3  | 34956   | 12016 | ?     | Ss   | 16:09 | 0:00 | /usr/lib/syst    |
| root     | 150 | 0.0  | 0.0  | 2880    | 1916  | ?     | Ss   | 16:09 | 0:00 | /bin/sh /usr/    |
| root     | 151 | 0.0  | 0.0  | 4464    | 3024  | ?     | Ss   | 16:09 | 0:00 | /usr/sbin/cro    |
| message+ | 152 | 0.0  | 0.1  | 8772    | 5176  | ?     | Ss   | 16:09 | 0:00 | @dbus-daemon     |
| root     | 156 | 0.3  | 0.7  | 44076   | 29388 | ?     | Ss   | 16:09 | 0:00 | /usr/bin/pyth    |
| root     | 161 | 0.1  | 0.2  | 18420   | 9240  | ?     | Ss   | 16:09 | 0:00 | /usr/lib/syst    |
| root     | 165 | 0.1  | 0.4  | 1792292 | 15520 | ?     | Ssl  | 16:09 | 0:00 | /usr/libexec/    |
| syslog   | 181 | 0.0  | 0.1  | 220532  | 5396  | ?     | Ssl  | 16:09 | 0:00 | /usr/sbin/rsy    |
| root     | 183 | 0.0  | 0.2  | 10724   | 7568  | ?     | Ss   | 16:09 | 0:00 | sshd: /usr/sb    |
| _chrony  | 213 | 0.0  | 0.1  | 20404   | 6148  | ?     | S    | 16:09 | 0:00 | /usr/sbin/chr    |
| _chrony  | 227 | 0.0  | 0.0  | 12076   | 2408  | ?     | S    | 16:09 | 0:00 | /usr/sbin/chr    |
| root     | 231 | 0.2  | 0.8  | 123444  | 32164 | ?     | Ssl  | 16:09 | 0:00 | /usr/bin/pyth    |
| root     | 240 | 0.0  | 0.0  | 5484    | 2664  | tty1  | Ss+  | 16:09 | 0:00 | /usr/sbin/age    |
| root     | 334 | 0.0  | 0.0  | 3176    | 1116  | ?     | Ss   | 16:09 | 0:00 | /init            |
| root     | 335 | 0.0  | 0.0  | 3192    | 1252  | ?     | S    | 16:09 | 0:00 | /init            |
| praveen  | 336 | 0.0  | 0.1  | 6260    | 5508  | pts/0 | Ss   | 16:09 | 0:00 | -bash            |
| root     | 338 | 0.0  | 0.1  | 8632    | 5472  | ?     | Ss   | 16:09 | 0:00 | login -- praveen |
| praveen  | 383 | 0.0  | 0.3  | 22728   | 13112 | ?     | Ss   | 16:09 | 0:00 | /usr/lib/syst    |
| praveen  | 385 | 0.0  | 0.1  | 22892   | 4068  | ?     | S    | 16:09 | 0:00 | (sd-pam)         |
| praveen  | 411 | 0.0  | 0.1  | 6128    | 5352  | pts/1 | Ss+  | 16:09 | 0:00 | -bash            |
| praveen  | 525 | 0.0  | 0.1  | 7152    | 4236  | pts/0 | R+   | 16:11 | 0:00 | ps aux           |

praveen@praveenasus:~\$ ps -ef



```
praveen@praveenasus: ~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0  16:09 ?        00:00:01 /sbin/init
root           2          1  0  16:09 hvc0     00:00:00 /init
root           6          2  0  16:09 hvc0     00:00:00 plan9 --control-socket 7 -
root          50          1  0  16:09 ?        00:00:00 /usr/lib/systemd/systemd-j
systemd+      85          1  0  16:09 ?        00:00:00 /usr/lib/systemd/systemd-r
root          92          1  0  16:09 ?        00:00:00 /usr/lib/systemd/systemd-u
root         150          1  0  16:09 ?        00:00:00 /bin/sh /usr/lib/systemd/s
root         151          1  0  16:09 ?        00:00:00 /usr/sbin/cron -f -P
message+     152          1  0  16:09 ?        00:00:00 @dbus-daemon --system --ad
root         156          1  0  16:09 ?        00:00:00 /usr/bin/python3 /usr/bin/
root         161          1  0  16:09 ?        00:00:00 /usr/lib/systemd/systemd-l
root         165          1  0  16:09 ?        00:00:00 /usr/libexec/wsl-pro-servi
syslog       181          1  0  16:09 ?        00:00:00 /usr/sbin/rsyslogd -n -iNO
root         183          1  0  16:09 ?        00:00:00 sshd: /usr/sbin/sshd -D [l
_chrony      213        150  0  16:09 ?        00:00:00 /usr/sbin/chronyd -n -F 1
_chrony      227        213  0  16:09 ?        00:00:00 /usr/sbin/chronyd -n -F 1
root         231          1  0  16:09 ?        00:00:00 /usr/bin/python3 /usr/shar
root         240          1  0  16:09 tty1     00:00:00 /usr/sbin/agetty --noreset
root         334          2  0  16:09 ?        00:00:00 /init
root         335        334  0  16:09 ?        00:00:00 /init
praveen      336        335  0  16:09 pts/0    00:00:00 -bash
root         338          2  0  16:09 ?        00:00:00 login -- praveen
praveen      383          1  0  16:09 ?        00:00:00 /usr/lib/systemd/systemd -
praveen      385        383  0  16:09 ?        00:00:00 (sd-pam)
praveen      411        338  0  16:09 pts/1    00:00:00 -bash
praveen      624        336  0  16:15 pts/0    00:00:00 ps -ef
praveen@praveenasus: ~$
```

```
praveen@praveenasus: ~$ ps -u root
```

| PID | TTY  | TIME     | CMD             |
|-----|------|----------|-----------------|
| 1   | ?    | 00:00:01 | systemd         |
| 2   | hvc0 | 00:00:00 | init-systemd(Ub |
| 6   | hvc0 | 00:00:00 | init            |
| 50  | ?    | 00:00:00 | systemd-journal |
| 92  | ?    | 00:00:00 | systemd-udevd   |
| 150 | ?    | 00:00:00 | chronyd-starter |
| 151 | ?    | 00:00:00 | cron            |
| 156 | ?    | 00:00:00 | networkd-dispat |
| 161 | ?    | 00:00:00 | systemd-logind  |
| 165 | ?    | 00:00:00 | wsl-pro-service |
| 183 | ?    | 00:00:00 | sshd            |
| 231 | ?    | 00:00:00 | unattended-upgr |
| 240 | tty1 | 00:00:00 | agetty          |
| 334 | ?    | 00:00:00 | SessionLeader   |
| 335 | ?    | 00:00:00 | Relay(336)      |
| 338 | ?    | 00:00:00 | login           |

```
praveen@praveenasus: ~$
```

## ► System Process

### Process Name: systemd

- systemd is the first process started during Linux boot.
- It manages services and background processes in the system.

## ► User Process

### Process Name: bash

- bash is a command-line shell used by users.

- It allows users to interact with the Linux operating system through commands.

### ► Explanation of Columns

#### •PID (Process ID)

A unique number assigned to every running process.

#### •TTY (Terminal Type)

Indicates the terminal associated with the process. A question mark (?) means the process is not attached to a terminal.

#### •STAT (Process State)

Shows the current state of the process.

- R = Running
- S = Sleeping
- Z = Zombie
- T = Stopped

#### •TIME

Displays the total CPU time consumed by the process since it started.

#### •CMD

Shows the command or program that started the process.

### Task – 4: Process Monitoring Comparison

#### SOLUTION-

### 1. ps aux

The ps aux command displays detailed information about all processes.

### 2. top

The top command provides real-time monitoring of processes and system resources.

### 3. htop

the htop services installed using **sudo apt install htop** command

The htop command is an advanced interactive process viewer.

```
praveen@praveenasus: ~$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.1  0.4  24560 15064 ?        Ss   16:09   0:01 /sbin/init
root           2  0.0  0.0   3172  2212 hvcd0    Sl+  16:09   0:00 /init
root           6  0.0  0.0   3172  2048 hvcd0    Sl+  16:09   0:00 plan9 --con
root          50  0.0  0.4  50380 16380 ?        S<s  16:09   0:00 /usr/lib/sy
systemd+      85  0.0  0.3  22524 14260 ?        Ss   16:09   0:00 /usr/lib/sy
root          92  0.1  0.3  34956 12016 ?        Ss   16:09   0:01 /usr/lib/sy
root         150  0.0  0.0   2880  1916 ?        Ss   16:09   0:00 /bin/sh /us
root         151  0.0  0.0   4464  3024 ?        Ss   16:09   0:00 /usr/sbin/c
message+     152  0.0  0.1   8772  5180 ?        Ss   16:09   0:00 @dbus-daemo
root         156  0.0  0.7  44076 29388 ?        Ss   16:09   0:00 /usr/bin/py
root         161  0.0  0.2  18420  9240 ?        Ss   16:09   0:00 /usr/lib/sy
syslog       181  0.0  0.1  220532  5396 ?        Ssl  16:09   0:00 /usr/sbin/r
root         183  0.0  0.2  10724  7568 ?        Ss   16:09   0:00 sshd: /usr/
_chrony      213  0.0  0.1  20404  6148 ?        S    16:09   0:00 /usr/sbin/c
_chrony      227  0.0  0.0  12076  2408 ?        S    16:09   0:00 /usr/sbin/c
root         231  0.0  0.8 123444 32164 ?        Ssl  16:09   0:00 /usr/bin/py
root         240  0.0  0.0   5484  2664 tty1     Ss+  16:09   0:00 /usr/sbin/a
root         334  0.0  0.0   3176  1116 ?        Ss   16:09   0:00 /init
root         335  0.0  0.0   3192  1252 ?        S    16:09   0:00 /init
praveen      336  0.0  0.1   6260  5508 pts/0    Ss   16:09   0:00 -bash
root         338  0.0  0.1   8632  5472 ?        Ss   16:09   0:00 login -- pr
praveen      383  0.0  0.3  22728 13176 ?        Ss   16:09   0:00 /usr/lib/sy
praveen      385  0.0  0.1  22892  4068 ?        S    16:09   0:00 (sd-pam)
praveen      411  0.0  0.1   6128  5352 pts/1    Ss+  16:09   0:00 -bash
praveen      862 100  0.1   7152  4232 pts/0    R+   16:34   0:00 ps aux
praveen@praveenasus: ~$
```



```
praveen@praveenasus:~$ top
```

```
top - 16:35:03 up 28 min, 1 user, load average: 0.04, 0.01, 0.00
```

```
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
```

```
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0
```

```
MiB Mem : 3621.1 total, 3089.6 free, 450.3 used, 156.7 buff/cache
```

```
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used. 3170.8 avail Mem
```

| PID | USER     | PR | NI | VIRT   | RES   | SHR   | S | %CPU | %MEM | TIME+   |
|-----|----------|----|----|--------|-------|-------|---|------|------|---------|
| 1   | root     | 20 | 0  | 24560  | 15064 | 11128 | S | 0.0  | 0.4  | 0:01.60 |
| 2   | root     | 20 | 0  | 3172   | 2212  | 2072  | S | 0.0  | 0.1  | 0:00.01 |
| 6   | root     | 20 | 0  | 3172   | 2048  | 1940  | S | 0.0  | 0.1  | 0:00.00 |
| 50  | root     | 19 | -1 | 50380  | 16380 | 15096 | S | 0.0  | 0.4  | 0:00.40 |
| 85  | systemd+ | 20 | 0  | 22524  | 14260 | 11648 | S | 0.0  | 0.4  | 0:00.18 |
| 92  | root     | 20 | 0  | 34956  | 12016 | 8944  | S | 0.0  | 0.3  | 0:01.75 |
| 150 | root     | 20 | 0  | 2880   | 1916  | 1792  | S | 0.0  | 0.1  | 0:00.13 |
| 151 | root     | 20 | 0  | 4464   | 3024  | 2776  | S | 0.0  | 0.1  | 0:00.03 |
| 152 | message+ | 20 | 0  | 8772   | 5180  | 4512  | S | 0.0  | 0.1  | 0:00.10 |
| 156 | root     | 20 | 0  | 44076  | 29388 | 14264 | S | 0.0  | 0.8  | 0:00.50 |
| 161 | root     | 20 | 0  | 18420  | 9240  | 8004  | S | 0.0  | 0.2  | 0:00.19 |
| 181 | syslog   | 20 | 0  | 220532 | 5396  | 4468  | S | 0.0  | 0.1  | 0:00.14 |
| 183 | root     | 20 | 0  | 10724  | 7568  | 6408  | S | 0.0  | 0.2  | 0:00.04 |
| 213 | _chrony  | 20 | 0  | 20404  | 6148  | 5376  | S | 0.0  | 0.2  | 0:00.10 |
| 227 | _chrony  | 20 | 0  | 12076  | 2408  | 1768  | S | 0.0  | 0.1  | 0:00.00 |
| 231 | root     | 20 | 0  | 123444 | 32164 | 17548 | S | 0.0  | 0.9  | 0:00.36 |
| 240 | root     | 20 | 0  | 5484   | 2664  | 2480  | S | 0.0  | 0.1  | 0:00.01 |
| 334 | root     | 20 | 0  | 3176   | 1116  | 980   | S | 0.0  | 0.0  | 0:00.00 |
| 335 | root     | 20 | 0  | 3192   | 1252  | 1108  | S | 0.0  | 0.0  | 0:00.01 |
| 336 | praveen  | 20 | 0  | 6260   | 5508  | 3840  | S | 0.0  | 0.1  | 0:00.13 |
| 338 | root     | 20 | 0  | 8632   | 5472  | 4636  | S | 0.0  | 0.1  | 0:00.02 |
| 383 | praveen  | 20 | 0  | 22728  | 13176 | 10740 | S | 0.0  | 0.4  | 0:00.17 |
| 385 | praveen  | 20 | 0  | 22892  | 4068  | 2072  | S | 0.0  | 0.1  | 0:00.00 |
| 411 | praveen  | 20 | 0  | 6128   | 5352  | 3808  | S | 0.0  | 0.1  | 0:00.03 |
| 878 | praveen  | 20 | 0  | 8160   | 5852  | 3748  | R | 0.0  | 0.2  | 0:00.02 |

```
praveen@praveenasus:~$ htop
```

```

praveen@praveenasus: ~
0[ | 0.4%] 3[ | 0.8%] 6[ 0.0%] 9[ 0.0%]
1[ 0.0%] 4[ 0.0%] 7[ 0.0%] 10[ 0.0%]
2[ 0.0%] 5[ 0.0%] 8[ 0.0%] 11[ 0.0%]
Mem[|||||] 390M/3.54G Tasks: 29, 12 thr, 0 kthr; 1 running
Swp[ 0K/1.00G] Load average: 0.03 0.02 0.00
Uptime: 00:30:31

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
1041 praveen 20 0 5776 4444 3500 R 0.4 0.1 0:00.21 htop
1 root 20 0 24560 15064 11128 S 0.0 0.4 0:01.63 /sbin/init
2 root 20 0 3172 2212 2072 S 0.0 0.1 0:00.01 /init
6 root 20 0 3172 2048 1940 S 0.0 0.1 0:00.00 plan9 --con
7 root 20 0 3172 2048 0 S 0.0 0.1 0:00.00 plan9 --con
8 root 20 0 3172 2212 0 S 0.0 0.1 0:00.00 /init
50 root 19 -1 50380 16456 15172 S 0.0 0.4 0:00.40 /usr/lib/sy
85 systemd-re 20 0 22524 14260 11648 S 0.0 0.4 0:00.18 /usr/lib/sy
92 root 20 0 34956 12016 8944 S 0.0 0.3 0:01.84 /usr/lib/sy
150 root 20 0 2880 1916 1792 S 0.0 0.1 0:00.13 /bin/sh /us
151 root 20 0 4464 3024 2776 S 0.0 0.1 0:00.03 /usr/sbin/c
152 messagebus 20 0 8772 5204 4512 S 0.0 0.1 0:00.12 @dbus-daemo
156 root 20 0 44076 29388 14264 S 0.0 0.8 0:00.50 /usr/bin/py
161 root 20 0 18420 9240 8004 S 0.0 0.2 0:00.20 /usr/lib/sy
181 syslog 20 0 215M 5396 4468 S 0.0 0.1 0:00.10 /usr/sbin/r
183 root 20 0 10724 7568 6408 S 0.0 0.2 0:00.04 sshd: /usr/
213 _chrony 20 0 20404 6148 5376 S 0.0 0.2 0:00.10 /usr/sbin/c
224 syslog 20 0 215M 5396 0 S 0.0 0.1 0:00.02 /usr/sbin/r
225 syslog 20 0 215M 5396 0 S 0.0 0.1 0:00.00 /usr/sbin/r
226 syslog 20 0 215M 5396 0 S 0.0 0.1 0:00.01 /usr/sbin/r
227 _chrony 20 0 12076 2408 1768 S 0.0 0.1 0:00.00 /usr/sbin/c
231 root 20 0 120M 32164 17548 S 0.0 0.9 0:00.36 /usr/bin/py
240 root 20 0 5484 2664 2480 S 0.0 0.1 0:00.01 /usr/sbin/a
290 root 20 0 120M 32164 0 S 0.0 0.9 0:00.00 /usr/bin/py
334 root 20 0 3176 1116 980 S 0.0 0.0 0:00.00 /init
335 root 20 0 3192 1252 1108 S 0.0 0.0 0:00.06 /init
336 praveen 20 0 6260 5508 3840 S 0.0 0.1 0:00.16 -bash
338 root 20 0 8632 5472 4636 S 0.0 0.1 0:00.02 login -- pr
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice -F8Nice +F9Kill F10Qui

```

## **Comparison of Commands**

### **ps aux**

- Provides a one-time snapshot of all running processes.
- Useful for scripting and quick process checks.
- Does not update automatically.

### **top**

- Displays real-time information about CPU, memory, and running processes.
- Updates continuously.
- Useful for monitoring system performance and troubleshooting.

### **htop**

- Interactive and user-friendly version of top.
- Provides color-coded output and easier navigation.
- Allows users to manage processes directly from the interface.